A System-Level Framework for Sentiment-Aware Reflective Writing Systems: Modeling Temporal Emotional Patterns with Interpretability and Ethical Safety

Gabu Sai Yamini Devi
Independent Researcher
saiyaminireddy@gmail.com

February, 2026

Abstract

Sentiment analysis has become a cornerstone of affective computing applications, enabling systems to detect emotional valence in user-generated text. However, when deployed in reflective writing and coaching contexts, sentiment-based systems face a fundamental limitation: sentiment polarity does not reliably indicate user intent, therapeutic need, or appropriate system response. This paper formally analyzes the gap between sentiment detection and intent interpretation in emotion-aware interactive systems. We define sentiment as a function $S : \mathcal{X} \to Y_s$ mapping text to polarity labels, and intent as a context-dependent function $I : \mathcal{X} \times \mathcal{C} \times \mathcal{G} \times \mathcal{H} \to \mathcal{A}$ incorporating conversational context, user goals, and interaction history. Through system-level observations from early-stage prototype deployment, we demonstrate that $S(x)$ alone is statistically insufficient for determining appropriate system actions. We present a multi-layer intent modeling framework that augments sentiment detection with pragmatic reasoning, temporal pattern recognition, and goal-aware interpretation. Our central contribution is a utility-theoretic framing of emotional response selection that explicitly models asymmetric costs (false intervention vs. missed support), derives optimal decision thresholds, and incorporates ethical constraints through action-space restriction.

Keywords: sentiment analysis, intent modeling, affective computing, reflective writing systems, pragmatic NLP, explainable AI, digital mental health, emotion recognition, dialogue policy learning, utility-theoretic decision making

1 Notation Table

Symbol	Definition
$\mathcal{X}$	Text input space
x, x_t	Text input (at time t)
Y_s	Sentiment label space $\{-1, 0, +1\}$
$S(x)$	Sentiment classifier function
$\mathcal{C}$	Conversational context
c, c_t	Specific context instance (at time t)
$\mathcal{G}$	User goal space
g, g_t	User goal (at time t)
$\mathcal{H}$	Temporal user history
h_t	Hidden state in temporal model
$\mathcal{A}$	System action space
a, a_t	System action (at time t)
$I(x, c, g, \mathcal{H})$	Intent function
z, z_t	Latent true user state (at time t)
$\mathcal{Z}$	Latent state space
$U(a, z)$	Utility function
θ	Model parameters
$L_1, \ldots, L_5$	Framework layers 1 through 5
$P(\cdot)$	Probability distribution
$\mathbb{E}[\cdot]$	Expectation operator
C_{fp}, C_{fn}	Cost of false positive, false negative

Table 1: Notation used throughout the paper.

2 Introduction

The proliferation of natural language processing (NLP) techniques in interactive systems has enabled unprecedented capabilities in understanding and responding to human emotional expression. Sentiment analysis, defined as the computational detection of affective states and opinions in text, has matured from simple lexicon-based methods to sophisticated transformer-based architectures achieving state-of-the-art performance across multiple benchmarks. However, as these technologies are increasingly deployed in sensitive domains such as mental health support, reflective journaling, and professional coaching, a critical gap emerges between what sentiment classifiers can detect and what users actually intend or need.

This paper addresses a fundamental question: When is sentiment analysis insufficient for guiding system behavior in emotion-aware applications? We argue that while sentiment detection is necessary for affective computing, it is not sufficient for systems that must interpret user intent, provide appropri-

ate interventions, or respect complex emotional dynamics. The distinction between detecting that text expresses negative sentiment and understanding whether negative sentiment indicates distress requiring intervention, productive self-criticism, or cathartic expression represents a challenge at the intersection of NLP, pragmatics, and human-computer interaction.

Consider a user writing in a digital journal: "I failed the interview completely. I'm such an idiot." A sentiment classifier would correctly identify strong negative polarity. However, the appropriate system response depends on factors orthogonal to sentiment: Is this catastrophizing requiring cognitive reframing? Is it realistic self-assessment following genuine poor performance? Is it productive self-criticism motivating improvement? Is it part of a concerning pattern of self-deprecation? Without access to conversational context, user goals, historical patterns, and pragmatic intent, sentiment alone cannot guide system behavior.

2.1 Motivation and Scope

Our analysis is motivated by observations from early-stage prototype deployment in the domain of sentiment-aware digital journaling. This context involves users expressing emotions during reflective activities, yet requires system responses that transcend simple sentiment-to-action mappings. This domain exemplifies a broader class of emotion-aware interactive systems where misinterpretation carries meaningful consequences for user well-being, trust, and system effectiveness.

The contributions of this paper include:

1. Formal proof sketch demonstrating that sentiment $S(x)$ is not a sufficient statistic for intent-aware action selection $I(x, \mathcal{C}, \mathcal{G}, \mathcal{H})$.

2. Utility-theoretic framing of emotional response selection with explicit asymmetric cost modeling (C_{fp} vs. C_{fn}) and derived optimal decision thresholds.

3. Formalization of ethical constraints through action-space restriction and privacy-autonomy cost terms in the utility function.

4. Multi-layer intent modeling framework integrating sentiment analysis with pragmatic reasoning, temporal pattern recognition, and goal-aware interpretation.

5. System-level observations documenting failure modes when sentiment-only approaches are applied to reflective writing contexts.

6. Explicit positioning relative to dialogue policy learning, POMDP-based systems, and LLM-based approaches.

7. Concrete empirical validation strategy for future testing.

8. Identification of open research problems in interpretable emotion-aware systems.

2.2 Paper Organization

Section 2 reviews related work in sentiment analysis, affective computing, dialogue policy learning, and pragmatic NLP, including comparison to LLM-based approaches. Section 3 provides formal problem formalization. Section 4 analyzes failure modes of sentiment-based systems. Section 5 presents observations from early-stage prototype deployment. Section 6 introduces our proposed multi-layer framework. Section 7 provides mathematical formulation with decision-theoretic framework. Section 8 outlines research agenda including empirical validation strategy. Section 9 discusses ethical considerations. Section 10 concludes with explicit positioning of contributions.

3 Background and Related Work

3.1 Sentiment Analysis and Affective Computing

Sentiment analysis has evolved from early lexicon-based approaches to modern deep learning architectures. Zhang et al. [15] provide a comprehensive evaluation of large language models (LLMs) on sentiment analysis tasks, revealing that while LLMs excel at simple polarity classification, they struggle with nuanced tasks requiring understanding of specific sentiment phenomena, particularly aspect-based sentiment analysis and structured sentiment information.

The field of affective computing, pioneered by Rosalind W. Picard [11], encompasses systems that detect, interpret, and respond to human emotions. Ma et al. [9] examined generative technology for emotion recognition through a scoping review analyzing over 330 research papers, highlighting advances in autoencoders, GANs, diffusion models, and LLMs for emotion recognition across multiple modalities. However, most affective computing research focuses on emotion detection rather than the subsequent challenge of appropriate system response based on detected emotions.

3.2 Dialogue Policy Learning and User Modeling

Our work builds upon research in dialogue systems and policy learning. Traditional dialogue policy learning approaches use reinforcement learning to optimize system actions based on dialogue state [14]. POMDP-based dialogue systems [13] explicitly model uncertainty in dialogue state. However, these approaches typically operate in task-oriented settings with well-defined goals (e.g., restaurant booking, technical support) and objective success metrics

rather than open-domain emotional support contexts with subjective appropriateness criteria.

Recent work on reinforcement learning from human feedback (RLHF) demonstrates the value of learning from human preferences rather than predefined reward functions [2, 10]. Our framework extends these ideas to emotion-aware systems where appropriate responses depend on context, goals, and temporal patterns rather than simple task completion metrics.

User modeling in adaptive systems [1] addresses personalization based on user characteristics and interaction history. Adaptive tutoring systems [12] adjust instruction based on student knowledge state. Our work differs in focusing specifically on emotional expression in reflective contexts, where the relationship between observed behavior (sentiment) and appropriate response is mediated by complex pragmatic and therapeutic considerations.

3.3 Large Language Models for Contextual Reasoning

Recent large language models demonstrate impressive capabilities in contextual reasoning and can implicitly model some aspects of pragmatics, user goals, and temporal dynamics when prompted appropriately. However, LLM-based approaches face several limitations in safety-critical emotional support contexts:

1. **Uncertainty Calibration:** LLMs often exhibit poor calibration, expressing high confidence in incorrect outputs, making it difficult to distinguish reliable from unreliable responses in high-stakes scenarios.

2. **Explicit Utility Modeling:** LLMs lack explicit frameworks for modeling asymmetric costs (e.g., C_{fp} vs. C_{fn}). While they can be prompted to "be careful," they cannot formally optimize decision thresholds based on quantified costs.

3. **Transparent Cost Thresholds:** The decision-making process in LLMs is opaque. Users and developers cannot inspect or adjust the implicit tradeoffs the model makes between over-intervention and under-support.

4. **Enforceable Ethical Constraints:** Hard constraints (e.g., never diagnose, never prescribe medication) are difficult to guarantee in LLM outputs. Prompt-based constraints can be circumvented or ignored in edge cases.

5. **Interpretability:** While LLMs can generate explanations, these are post-hoc rationalizations rather than transparent reasoning traces. Our layered framework provides mechanistic interpretability through explicit intermediate representations.

Our framework complements LLM capabilities by providing explicit structure for utility-based decision making, enforceable ethical constraints, and transparent reasoning that can be inspected, validated, and corrected by users and developers.

3.4 Clinical NLP and Digital Mental Health

Clinical NLP research addresses mental health concerns through computational methods, including suicide risk detection [3] and depression identification [4]. However, most clinical NLP work focuses on detection and classification rather than the subsequent challenge of appropriate system response in interactive contexts.

Digital mental health interventions (DMHIs) increasingly incorporate emotion recognition to personalize support [7]. However, the application of sentiment analysis in mental health contexts raises critical questions about appropriate interpretation and response. A user expressing sadness may be engaging in healthy emotional processing, experiencing clinical depression, or simply having a bad day—distinctions that sentiment polarity alone cannot resolve.

3.5 Pragmatics and Intent Detection

Pragmatics concerns how context contributes to meaning beyond literal semantic content [8]. In NLP, pragmatic processing involves understanding implicit intentions, recognizing sarcasm, detecting presupposition, and inferring speaker goals. Intent detection, a core task in conversational AI, involves determining the underlying purpose behind user utterances.

Recent work examines how LLMs handle context-dependent meaning, implicature, and social norms. However, most intent detection research focuses on task-oriented dialogues with predefined intent taxonomies rather than open-domain emotional expression in reflective contexts.

3.6 Limitations of Existing Approaches and Our Contributions

While significant progress has been made in dialogue systems, affective computing, and clinical NLP, several gaps remain that our work addresses:

1. **Task-Oriented Focus:** Most dialogue policy learning operates in task-completion settings. Our work addresses open-ended reflective writing without predefined task goals.

2. **Detection Emphasis:** Affective computing emphasizes detection. We formalize the response selection problem with explicit utility functions.

3. **Implicit Cost Modeling:** Existing work rarely formalizes asymmetric intervention costs. We derive optimal thresholds from C_{fp} and C_{fn}.

4. **Limited Interpretability:** Deep RL and LLM approaches lack transparency. Our layered architecture provides mechanistic interpretability.

5. **Informal Ethical Constraints:** Ethics are often discussed qualitatively. We formalize constraints through action-space restriction and utility penalties.

Our novel contributions lie in: (1) formal insufficiency proof for sentiment in reflective contexts, (2) utility-threshold derivation under asymmetric ethical costs, (3) explicit autonomy-privacy tradeoff modeling, and (4) application to open-ended reflective writing rather than task completion.

3.7 Explainability and Ethics in Emotion AI

Recent attention to explainable AI (XAI) emphasizes the importance of interpretable models, particularly in high-stakes domains. The European Union's AI Act (Regulation EU 2024/1689) [6], enacted in 2024, explicitly addresses emotion recognition technologies within a risk-based regulatory framework. Ethical considerations include privacy, consent, emotional manipulation, and the risk of pathologizing normal emotional variation.

4 Problem Formalization

We now formally define the core concepts underlying our analysis.

Definition 1 (Text Input Space)**.** *Let $\mathcal{X}$ denote the space of all possible text inputs, where each $x \in \mathcal{X}$ is a sequence of tokens $x = (w_1, w_2, \ldots, w_n)$ with w_i drawn from vocabulary $\mathcal{V}$.*

Definition 2 (Sentiment Classifier)**.** *A sentiment classifier is a function:*

$$S : \mathcal{X} \to Y_s \tag{1}$$

where $Y_s = \{-1, 0, +1\}$ represents negative, neutral, and positive sentiment respectively. In practice, S may output a probability distribution $P(y_s \mid x)$ over sentiment labels.

Modern sentiment classifiers are typically implemented as neural networks f_θ with parameters θ:

$$S(x) = \arg\max_{y \in Y_s} f_\theta(x, y) \tag{2}$$

Definition 3 (Conversational Context)**.** *Let $\mathcal{C}$ represent the conversational context, including recent dialogue history, turn-level metadata (timestamps, interaction mode), and domain-specific context (e.g., daily journaling, therapeutic writing).*

Definition 4 (User Goals)**.** *Let $\mathcal{G}$ denote the space of user goals, which may include:*

- *Explicit goals: stated objectives (e.g., "emotional self-awareness")*
- *Implicit goals: inferred objectives (e.g., emotional catharsis)*
- *Meta-goals: goals about goals (e.g., "understand my emotional patterns")*

Definition 5 (Temporal User History)**.** *Let $\mathcal{H}$ represent the user's complete temporal interaction history:*

$$\mathcal{H} = \{(x_i, t_i, s_i, m_i)\}_{i=1}^{T} \tag{3}$$

where x_i is text, t_i is timestamp, s_i is detected sentiment, and m_i is metadata.

Definition 6 (Action Space and Utility)**.** *Let $\mathcal{A}$ represent the space of system actions (e.g., supportive response, reflective question, intervention alert, neutral acknowledgment). We define utility function:*

$$U : \mathcal{A} \times \mathcal{Z} \to \mathbb{R} \tag{4}$$

where $\mathcal{Z}$ represents the true user state (including needs, goals, mental health status). Higher utility indicates more appropriate actions.

Definition 7 (Intent Function - Discriminative Formulation)**.** *The intent function computes a probability distribution over appropriate actions:*

$$I : \mathcal{X} \times \mathcal{C} \times \mathcal{G} \times \mathcal{H} \to \Delta(\mathcal{A}) \tag{5}$$

where $\Delta(\mathcal{A})$ is the probability simplex over actions. The optimal action is:

$$a^* = \arg\max_{a \in \mathcal{A}} \mathbb{E}_{z \sim P(\cdot|x,c,g,\mathcal{H})}[U(a, z)] \tag{6}$$

The intent function integrates observable information $(x, c, g, \mathcal{H})$ to estimate which action maximizes expected utility given uncertainty about true user state z.

4.1 The Central Problem

The core problem we address is that sentiment classifiers provide only:

$$S : \mathcal{X} \to Y_s \tag{7}$$

while appropriate system behavior requires:

$$I : \mathcal{X} \times \mathcal{C} \times \mathcal{G} \times \mathcal{H} \to \Delta(\mathcal{A}) \tag{8}$$

Proposition 1 (Statistical Insufficiency of Sentiment for Intent)**.** *Sentiment $S(x)$ is not a sufficient statistic for intent-aware action selection. Formally, there exists $x \in \mathcal{X}$ and distinct contexts $(c_1, g_1, \mathcal{H}_1), (c_2, g_2, \mathcal{H}_2)$ such that:*

$$P(a^* \mid x, c_1, g_1, \mathcal{H}_1) \neq P(a^* \mid x, c_2, g_2, \mathcal{H}_2) \text{ even when } S(x) \text{ is constant} \tag{9}$$

Equivalently, sentiment is insufficient because:

$$P(a^* \mid x, c, g, \mathcal{H}) \neq P(a^* \mid S(x)) \tag{10}$$

Proof Sketch. We demonstrate existence by constructing specific cases where identical sentiment requires different actions.

Consider $x =$ "I'm feeling really down today" with $S(x) = -1$ (negative).

Case 1: User has been consistently negative for weeks ($\mathcal{H}_1$ shows declining trend), goal is daily journaling for self-awareness (g_1), context indicates isolation (c_1). Optimal action: $a_1^* =$ suggest professional support.

Case 2: User typically maintains positive sentiment but is reacting to specific recent event ($\mathcal{H}_2$ shows stable baseline with acute spike), goal is emotional processing (g_2), context indicates transient stressor (c_2). Optimal action: $a_2^* =$ empathetic acknowledgment.

Case 3: User is engaging in trauma-focused writing ($\mathcal{H}_3$ shows expected therapeutic decline), goal is processing difficult memories (g_3), context indicates therapist-guided practice (c_3). Optimal action: $a_3^* =$ validation without intervention.

The sentiment $S(x) = -1$ is constant across all three cases, yet the optimal actions differ: $a_1^* \neq a_2^* \neq a_3^*$. Therefore:

$$P(a^* \mid x, c_1, g_1, \mathcal{H}_1) \neq P(a^* \mid x, c_2, g_2, \mathcal{H}_2) \neq P(a^* \mid x, c_3, g_3, \mathcal{H}_3) \tag{11}$$

By constructing these distinct contexts with identical sentiment but different optimal actions, we show that $S(x)$ cannot be a sufficient statistic for I.

Intuitively, the mutual information between optimal action and full context exceeds that between action and sentiment alone:

$$I(A^*; X, \mathcal{C}, \mathcal{G}, \mathcal{H}) > I(A^*; S(X)) \tag{12}$$

This suggests that $(\mathcal{C}, \mathcal{G}, \mathcal{H})$ provide information about optimal actions beyond what sentiment captures, though formal proof of this inequality would require empirical measurement of these quantities. □

4.2 Decision-Theoretic Foundation

The system must select actions to maximize expected utility under uncertainty about true user state. The decision problem is:

$$a^* = \arg\max_{a \in \mathcal{A}} \mathbb{E}_{z \sim P(\cdot \mid x,c,g,\mathcal{H})}[U(a,z)] \tag{13}$$

where $P(z \mid x, c, g, \mathcal{H})$ is the system's belief about true user state given observable information, and $U(a, z)$ quantifies the quality of action a when true state is z.

The utility function $U(a, z)$ incorporates multiple considerations:

$$U(a,z) = w_1 \cdot \text{Appropriateness}(a,z) + w_2 \cdot \text{Safety}(a,z) - \lambda_1 \cdot \text{Privacy_Cost}(a,z) - \lambda_2 \cdot \text{Autonomy_Cost}(a,z \tag{14}$$

where weights $w_1, w_2, \lambda_1, \lambda_2$ encode relative importance of different objectives.

5 Failure Modes of Sentiment-Based Systems

We identify several systematic failure modes when sentiment analysis is used as the sole basis for system behavior in reflective and coaching contexts.

5.1 Context Collapse

Definition: Context collapse occurs when the system treats temporally or situationally isolated expressions as requiring immediate response, ignoring broader patterns in history $\mathcal{H}$.

Example: A user writes "worst day ever" after a minor inconvenience. A sentiment-only system might trigger unnecessary intervention, while context-aware analysis would recognize this as hyperbolic expression inconsistent with concerning patterns.

Mathematical Formulation: Context collapse occurs when the system computes:

$$P(a \mid x, c, g, \mathcal{H}) \approx P(a \mid S(x)) \tag{15}$$

rather than properly incorporating historical context as in Equation 13.

The information loss can be quantified as:

$$\Delta I = I(A^*; S(x), \mathcal{H}) - I(A^*; S(x)) > 0 \tag{16}$$

where ΔI represents bits of information about optimal actions contained in history $\mathcal{H}$ beyond sentiment.

5.2 Therapeutic Misalignment

Definition: Therapeutic misalignment occurs when system responses contradict established therapeutic principles or user goals.

Example: User engaging in cognitive restructuring writes: "I thought I was worthless, but I realize that's not true." The sentence contains negative sentiment ("worthless"), but represents therapeutic progress. A sentiment-only system might offer unnecessary support, undermining the user's agency.

Formal Characterization: Given user goal $g \in \mathcal{G}$ (e.g., "develop self-compassion"), therapeutic misalignment occurs when:

$$a_{\text{system}} \notin \text{Align}(g) \tag{17}$$

where $\text{Align}(g) \subseteq \mathcal{A}$ is the set of therapeutically appropriate actions for goal g.

The cost of misalignment can be expressed through utility:

$$\text{Loss} = U(a_{\text{appropriate}}, z) - U(a_{\text{actual}}, z) > 0 \tag{18}$$

where $a_{\text{appropriate}} \in \text{Align}(g)$ and $a_{\text{actual}} \notin \text{Align}(g)$.

5.3 Intent Ambiguity Under Negative Sentiment

Multiple distinct intents can produce identical sentiment signatures:

- Venting: Cathartic expression without need for intervention
- Help-seeking: Implicit request for support or resources
- Self-criticism: Productive reflection vs. harmful rumination
- Situational response: Transient reaction to specific event
- Clinical concern: Pattern indicating potential mental health need

For negative sentiment, the conditional entropy of intent given sentiment is high:

$$H(\text{Intent} \mid S(x) = -1) \gg 0 \tag{19}$$

necessitating disambiguation through context, goals, and history.

5.4 Privacy-Invasive Over-Inference

Given sentiment history $\{s_i\}_{i=1}^{T}$ and inference function $\Phi : \{s_i\} \to \mathcal{D}$ where $\mathcal{D}$ represents mental health diagnoses or risk categories, over-inference occurs when the system makes inferences beyond user consent or clinical validity.

The ethical violation can be quantified through utility penalty:

$$U_{\text{ethical}} = U_{\text{task}} - \lambda_{\text{privacy}} \times \max(0, I(\text{Diagnosis}; \text{System_Output}) - I_{\text{consented}}) \tag{20}$$

5.5 Temporal Pattern Misinterpretation

Given sentiment time series $\{s_t\}_{t=1}^{T}$ with trend $\nabla s < 0$, appropriate response depends on latent causal factors. A user showing declining sentiment over weeks could indicate progressive deterioration, increased emotional honesty, external stressor, or therapeutic processing—states requiring different responses despite identical sentiment trends.

6 Observations from Early-Stage Prototype Deployment

We present observations from a prototype system evaluated in limited pilot testing. These qualitative observations motivated the theoretical framework presented in this paper rather than serving as empirical validation; formal validation would require the controlled study outlined in Section 9.9.

6.1 System Description

A sentiment-aware digital journaling application that tracks emotional patterns over time, providing users with visualizations of sentiment trends and optional prompts for deeper reflection. The system was deployed in early-stage testing to support users in reflective writing and emotional self-awareness practices.

6.2 Observed Challenge 1: Cathartic Expression

Problem: Users reported that the system sometimes offered intervention suggestions during cathartic venting. One user wrote: "I just needed to get my anger out. The system suggesting coping strategies felt like it was telling me my feelings were wrong."

Analysis: The system needed to distinguish:

$$\text{Catharsis:} \quad S(x) < 0 \wedge \text{Goal} = \text{EmotionalRelease} \wedge \neg\text{Help-seeking} \tag{21}$$

$$\text{DistressCall:} \quad S(x) < 0 \wedge \text{Goal} = \text{SeekingSupport} \wedge \text{Help-seeking} \tag{22}$$

Pragmatic markers help distinguish these:

- Catharsis: Exclamatory tone, metaphorical language, conclusion-oriented
- Distress call: Question-oriented, help-seeking language, uncertainty markers

Utility Consideration:

$$U(\text{Validation} \mid \text{catharsis}) = +10 \tag{23}$$

$$U(\text{Intervention} \mid \text{catharsis}) = -8 \quad \text{(interrupts healthy process)} \tag{24}$$

$$U(\text{Validation} \mid \text{distress}) = -5 \quad \text{(misses needed support)} \tag{25}$$

$$U(\text{SupportiveResponse} \mid \text{distress}) = +10 \tag{26}$$

System Limitation: Pure sentiment classification ($S(x) < 0$) triggered intervention for both catharsis and distress, resulting in inappropriate responses that users experienced as invalidating their healthy emotional expression.

6.3 Observed Challenge 2: Progress Through Negativity

Problem: Users engaging in therapeutic practices (e.g., writing about difficult past experiences) would see declining sentiment scores marked as concerning trends, despite this representing healthy processing.

Example: A user practicing trauma-focused writing saw sentiment decline over weeks as they addressed progressively more difficult memories. The system flagged this as concerning, while the user's therapist confirmed it represented expected therapeutic progress.

Formal Issue: Let g_{therapy} = "process difficult memories" and let $\text{Progress}_{\text{therapy}}(t)$ measure actual therapeutic progress at time t.

The system observed:

$$\frac{\partial S(x_t)}{\partial t} < 0 \quad \text{(declining sentiment)} \tag{27}$$

While the true relationship was:

$$\frac{\partial \text{Progress}_{\text{therapy}}(t)}{\partial t} > 0 \quad \text{(improving therapeutic state)} \tag{28}$$

A sentiment-only system incorrectly interprets declining $S(x_t)$ as deterioration. Appropriate interpretation requires understanding:

1. User goal g_{therapy}
2. Domain knowledge: trauma processing typically involves temporarily increased negative affect
3. Expected pattern: sentiment decline followed by stabilization indicates progress

Decision-Theoretic Formulation: The system must decide between:

- a_1: Flag as concerning, suggest intervention (Cost if false positive: $C_{fp} = 8$)

- a_2: Recognize as therapeutic progress, provide validation (Cost if false negative: $C_{fn} = 15$)

With uncertain state (therapeutic progress vs. deterioration), optimal decision:

$$\text{Choose } a_1 \text{ if } P(\text{deterioration} \mid \text{trend}, g, \mathcal{H}) > \frac{C_{fp}}{C_{fp} + C_{fn}} \approx 0.35 \tag{29}$$

With proper goal and history modeling, $P(\text{deterioration} \mid \text{trauma-processing-goal}, \text{expected-decline-pattern}, \mathcal{H}) < 0.35$, so system correctly selects validation.

6.4 Observed Challenge 3: Individual Baseline Variation

Problem: The system used population-level sentiment norms to identify concerning patterns, failing to account for individual differences in baseline emotional expression.

Example: A user with naturally reserved emotional expression (baseline sentiment ≈ 0) wrote an entry with sentiment $+0.3$. For this user, this represented unusually positive affect. However, the system classified it as "neutral" based on population norms, missing an opportunity to acknowledge the user's positive state.

Conversely, a user with typically exuberant expression (baseline sentiment $\approx +0.8$) wrote an entry with sentiment $+0.4$. While still "positive" by population norms, this represented a significant decline from their personal baseline, yet the system provided no support.

Formal Characterization: Let μ_{pop} be population mean sentiment and μ_{user} be individual user's baseline. Appropriate response depends on:

$$\Delta_{\text{personal}} = s_t - \mu_{\text{user}} \quad \text{(deviation from personal baseline)} \tag{30}$$

rather than:

$$\Delta_{\text{population}} = s_t - \mu_{\text{pop}} \quad \text{(deviation from population mean)} \tag{31}$$

A sentiment-only system uses $\Delta_{\text{population}}$, while an intent-aware system should compute Δ_{personal} from history $\mathcal{H}$.

6.5 Common Patterns Across Observations

All three challenges reveal:

1. **Goal-Dependence:** Appropriate interpretation depends critically on user goals

2. **Temporal Context:** Single-point sentiment is insufficient; patterns matter

3. **Personalization:** Individual baselines vary significantly
4. **Pragmatic Signals:** Surface-level pragmatics disambiguate intent
5. **Cost Asymmetry:** Costs of false/missed intervention are context-dependent

These observations motivated the multi-layer intent modeling framework presented in Section 7.

7 Proposed Multi-Layer Intent Modeling Framework

We propose a framework that augments sentiment detection with additional layers of analysis, enabling intent-aware system behavior while maintaining interpretability.

7.1 Framework Architecture

Our framework consists of five layers (see Figure 1):

TEXT INPUT: x

$\downarrow$

LAYER 1: Sentiment Detection

$S(x) \to P(y_s \mid x)$ where $y_s \in \{-1, 0, +1\}$

Output: Sentiment score with confidence

$\downarrow$

LAYER 2: Pragmatic Analysis (uses Context $\mathcal{C}$)

$\text{Pragma}(x, c) \to \{\text{assertion, question, expression}\}$

Output: Speech act classification

$\downarrow$

LAYER 3: Temporal Pattern Recognition (uses History $\mathcal{H}$)

$\text{Pattern}(\mathcal{H}) \to \{\text{stable, declining, improving}\}$

Output: Temporal trend with attention weights

$\downarrow$

LAYER 4: Goal Alignment (uses Goals $\mathcal{G}$)

$\text{GoalAlign}(s, \text{pattern}, g) \to \{\text{aligned, misaligned}\}$

Output: Compatibility score with user goals

$\downarrow$

LAYER 5: Utility-Based Action Selection

$a^* = \arg\max_a \mathbb{E}[U(a, z) \mid L_1, L_2, L_3, L_4]$

Output: Optimal action with explanation

$\downarrow$

SYSTEM RESPONSE: a^*

Figure 1: Multi-layer intent modeling framework architecture. Each layer produces interpretable outputs that feed into utility-based action selection.

Layer 1 - Sentiment Detection: Standard sentiment classification providing base affective signal:

$$L_1 : S(x) \to P(y_s \mid x), \quad y_s \in \{-1, 0, +1\} \tag{32}$$

Layer 2 - Pragmatic Analysis: Contextual interpretation incorporating discourse markers and speech acts:

$$L_2 : \text{Pragma}(x, c) \to \{\text{assertion}, \text{question}, \text{expression}, \text{request}, \ldots\} \tag{33}$$

Layer 3 - Temporal Pattern Recognition: Analysis of sentiment dynamics over time using recurrent model:

$$L_3 : \text{Pattern}(\mathcal{H}) \to \{\text{stable}, \text{declining}, \text{improving}, \text{cyclical}, \ldots\} \tag{34}$$

Layer 4 - Goal-Awareness Module: Maps detected patterns to user goals:

$$L_4 : \text{GoalAlign}(s, \text{pattern}, g) \to \{\text{aligned}, \text{misaligned}, \text{ambiguous}\} \tag{35}$$

Layer 5 - Utility-Based Action Selection: Integrates all layers to maximize expected utility:

$$L_5 : a^* = \arg\max_{a \in \mathcal{A}} \mathbb{E}[U(a, z) \mid L_1(x), L_2(x, c), L_3(\mathcal{H}), L_4(s, \text{pattern}, g)] \tag{36}$$

7.2 Toy Example: Numeric Utility Tradeoff

Consider a user message: "I feel terrible today"

Layer Outputs:

- L_1: Sentiment $= -0.8$ (strongly negative), Confidence $= 0.92$
- L_2: Pragmatic = expression (not question, not help-seeking)
- L_3: Pattern = stable_baseline (no concerning trend in $\mathcal{H}$)
- L_4: Goal alignment = aligned (goal: "daily emotional check-in")

Candidate Actions with Utility Estimates:

Action a_1: Validation without intervention

"Thank you for sharing. It's okay to have difficult days."

$$U(a_1 \mid z_1 : \text{healthy_expression}) = +10 \tag{37}$$

$$U(a_1 \mid z_2 : \text{distress_call}) = -5 \tag{38}$$

$$P(z_1 \mid \text{layers}) = 0.85, \quad P(z_2 \mid \text{layers}) = 0.15 \tag{39}$$

$$\mathbb{E}[U(a_1)] = 0.85 \times 10 + 0.15 \times (-5) = 7.75 \tag{40}$$

Action a_2: Supportive intervention

"I'm sorry you're feeling terrible. Would you like to talk about what's happening or see some coping resources?"

$$U(a_2 \mid z_1 : \text{healthy_expression}) = -3 \quad \text{(over-intervention)} \tag{41}$$

$$U(a_2 \mid z_2 : \text{distress_call}) = +12 \tag{42}$$

$$\mathbb{E}[U(a_2)] = 0.85 \times (-3) + 0.15 \times 12 = -0.75 \tag{43}$$

Action a_3: Reflective prompt
"What's contributing to how you're feeling today?"

$$U(a_3 \mid z_1 : \text{healthy_expression}) = +7 \tag{44}$$

$$U(a_3 \mid z_2 : \text{distress_call}) = +4 \tag{45}$$

$$\mathbb{E}[U(a_3)] = 0.85 \times 7 + 0.15 \times 4 = 6.55 \tag{46}$$

Decision: Select a_1 (Validation) with $\mathbb{E}[U] = 7.75$
This example demonstrates how the framework:

1. Integrates multiple layers to estimate $P(z \mid \text{layers})$
2. Computes expected utility for each action under uncertainty
3. Balances risks of over-intervention vs. under-support
4. Makes explainable decisions based on layer outputs

Without layers 2–4, a sentiment-only system would see $S(x) = -0.8$ and likely choose intervention (a_2), resulting in negative utility when state is healthy expression.

7.3 Interpretability Mechanisms

Each layer produces human-understandable outputs with confidence scores:

Example System Explanation to User:

"I noticed your message expressed difficult feelings (sentiment: -0.8, high confidence). Based on the way you phrased it—as a statement rather than a question—and your stable pattern over the past week, this seems like a normal check-in rather than a request for help. Your goal of daily emotional awareness supports acknowledging feelings without unnecessary intervention. I'm here if you'd like to explore further."

This explanation references specific layer outputs while maintaining natural language accessibility, enabling users to understand and potentially correct the system's reasoning.

8 Mathematical Formulation of Intent-Aware Modeling

We formalize intent-aware modeling as a discriminative prediction problem with utility-based decision making.

8.1 Discriminative Formulation

The system computes the probability distribution over appropriate actions:

$$P(a \mid x, c, g, \mathcal{H}; \theta) \tag{47}$$

We model this discriminatively, directly estimating action probabilities from observable information without assuming actions cause text. The discriminative approach models:

$$P(a \mid x, c, g, \mathcal{H}; \theta) = \text{softmax}(\Psi(a, x, c, g, \mathcal{H}; \theta)) \tag{48}$$

where Ψ is a scoring function with parameters θ.

For tractability and interpretability, we factor through our layered architecture:

$$P(a \mid x, c, g, \mathcal{H}; \theta) \approx P(a \mid s, p, \pi, q; \theta) \tag{49}$$

where:

- $s = L_1(x; \theta_1)$: sentiment
- $p = L_2(x, c; \theta_2)$: pragmatic category
- $\pi = L_3(\mathcal{H}; \theta_3)$: temporal pattern
- $q = L_4(s, \pi, g; \theta_4)$: goal alignment

Note: This conditional independence assumption states that intermediate representations (s, p, π, q) capture sufficient information for action selection. This is an approximation that introduces potential optimality loss but enables interpretability and modular training. The approximation error depends on how well the intermediate layers capture task-relevant information.

8.2 Factorized Model

We factorize the model through framework layers:

$$\begin{aligned} P(a \mid x, c, g, \mathcal{H}; \theta) = \sum_{s,p,\pi,q} & P(a \mid s, p, \pi, q; \theta_5) \times P(s \mid x; \theta_1) \\ & \times P(p \mid x, c; \theta_2) \times P(\pi \mid \mathcal{H}; \theta_3) \times P(q \mid s, \pi, g; \theta_4) \end{aligned} \tag{50}$$

where $\theta = \{\theta_1, \theta_2, \theta_3, \theta_4, \theta_5\}$ are layer-specific parameters.

8.3 Utility-Based Learning Objective

Given training data $\mathcal{D} = \{(x_i, c_i, g_i, \mathcal{H}_i, a_i^*, r_i)\}_{i=1}^N$ where a_i^* are appropriate actions and r_i are utility/reward values, we optimize:

$$\mathcal{L}(\theta) = -\sum_{i=1}^{N} r_i \times \log P(a_i^* \mid x_i, c_i, g_i, \mathcal{H}_i; \theta) + \lambda\Omega(\theta) \tag{51}$$

where $\Omega(\theta)$ encourages interpretability:

$$\Omega(\theta) = \alpha \sum_{\ell=1}^{5} \|\theta_\ell\|_1 + \beta \sum_{\text{similar contexts}} \|\Psi(a_1) - \Psi(a_2)\|^2 \tag{52}$$

The first term penalizes model complexity; the second penalizes inconsistent actions for similar contexts.

8.4 Temporal Modeling

For Layer 3, we model sentiment evolution using LSTM with hidden state h_t:

$$h_t = \text{LSTM}(s_t, h_{t-1}; \theta_3) \tag{53}$$

$$P(\pi_t \mid \mathcal{H}; \theta_3) = \text{softmax}(W_\pi h_t + b_\pi) \tag{54}$$

where $W_\pi, b_\pi \in \theta_3$.

For interpretability, we compute attention weights showing which historical points contribute to pattern classification:

$$\alpha_i = \frac{\exp(\text{score}(h_t, s_i))}{\sum_j \exp(\text{score}(h_t, s_j))} \tag{55}$$

Note: We use LSTM rather than Transformers for: (1) long sequences (months of journaling), (2) online processing (updating with new entries), (3) simpler interpretability through hidden states. Alternative architectures such as hierarchical Transformers may provide improved modeling capacity at the cost of increased computational complexity and reduced transparency. The choice involves tradeoffs between expressive power, computational efficiency, and interpretability.

8.5 Goal-Conditioned Action Selection

Given user goal $g \in \mathcal{G}$, we model goal-conditional action distribution:

$$P(a \mid s, p, \pi, g; \theta_5) = \text{softmax}(\phi(s, p, \pi, a, g; \theta_5)) \tag{56}$$

where ϕ is a compatibility function:

$$\phi(s, p, \pi, a, g; \theta_5) = \theta_5^T [\text{embed}(s) \oplus \text{embed}(p) \oplus \text{embed}(\pi) \oplus \text{embed}(a) \oplus \text{embed}(g)] \quad (57)$$

This learns which actions serve which goals given sentiment, pragmatics, and patterns.

8.6 Ethical Constraints Through Action Space Restriction

We incorporate hard constraints preventing harmful behaviors:

$$\mathcal{A}_{\text{allowed}}(x, \mathcal{H}) = \mathcal{A} \setminus \mathcal{A}_{\text{harmful}}(x, \mathcal{H}) \quad (58)$$

where $\mathcal{A}_{\text{harmful}}$ includes actions that:

1. Make clinical diagnoses without qualification
2. Suggest medication or medical interventions
3. Dismiss persistent negative patterns
4. Over-generalize from insufficient data
5. Violate privacy through excessive inference

The restricted action space ensures:

$$P(a \mid x, c, g, \mathcal{H}; \theta) = \begin{cases} \text{softmax}(\Psi(a, \ldots))/Z & \text{if } a \in \mathcal{A}_{\text{allowed}} \\ 0 & \text{if } a \in \mathcal{A}_{\text{harmful}} \end{cases} \quad (59)$$

where Z normalizes over allowed actions.

8.7 Complete Decision-Theoretic Framework and Utility Specification

The optimal action maximizes expected utility under uncertainty:

$$a^* = \arg \max_{a \in \mathcal{A}_{\text{allowed}}} \mathbb{E}_{z \sim P(\cdot \mid x, c, g, \mathcal{H}; \theta)} [U(a, z)] \quad (60)$$

where:

- $P(z \mid x, c, g, \mathcal{H}; \theta)$ is the system's belief about true user state
- $U(a, z)$ is the utility of action a when true state is z
- $\mathcal{A}_{\text{allowed}}$ enforces ethical constraints

The utility function decomposes as:

$$U(a, z) = w_{\text{task}} \cdot \text{Appropriateness}(a, z) + w_{\text{safety}} \cdot \text{Safety}(a, z) \\ - \lambda_{\text{privacy}} \cdot \text{Privacy_Cost}(a, z) - \lambda_{\text{autonomy}} \cdot \text{Autonomy_Cost}(a, z) \quad (61)$$

Weight Specification: The weights $\{w_{\text{task}}, w_{\text{safety}}, \lambda_{\text{privacy}}, \lambda_{\text{autonomy}}\}$ can be determined through multiple approaches:

1. **Expert Elicitation:** Mental health professionals and domain experts provide judgments on appropriate tradeoffs, which are then quantified through structured elicitation methods (e.g., analytic hierarchy process).

2. **User Personalization:** Individual users specify their preferences for system behavior (e.g., "I prefer minimal intervention" translates to higher $\lambda_{\text{autonomy}}$).

3. **Preference Learning:** Weights are learned from human feedback on system responses, similar to RLHF but with explicit utility structure enabling interpretability.

4. **Hybrid Approach:** Default weights from expert consensus, personalized through user interaction and feedback over time.

The decomposition into explicit components (appropriateness, safety, privacy, autonomy) enables transparent inspection and adjustment, contrasting with monolithic reward functions in standard RL or implicit objectives in LLMs.

Cost Formulation for Intervention Decisions: Let C_{fp} be the cost of false positive (intervening when unnecessary) and C_{fn} be the cost of false negative (missing needed intervention). The optimal decision threshold τ^* satisfies:

$$\tau^* = \frac{C_{fp}}{C_{fp} + C_{fn}} \quad (62)$$

The system intervenes when:

$$P(\text{need_intervention} \mid x, c, g, \mathcal{H}; \theta) > \tau^* \quad (63)$$

This makes explicit the tradeoff between over-intervention (undermining autonomy, cost C_{fp}) and under-intervention (missing support needs, cost C_{fn}).

Cost Determination: The costs C_{fp} and C_{fn} can be specified through:

- Expert judgment (clinical psychologists rate severity of each error type)

- User-reported impact (surveys measuring harm from inappropriate interventions)

- Empirical measurement (A/B testing to quantify downstream effects)

For the journaling system in Section 5.3 with expert-elicited costs $C_{fp} = 8$ and $C_{fn} = 15$:

$$\tau^* = \frac{8}{8 + 15} \approx 0.35 \tag{64}$$

The system only intervenes when belief that intervention is needed exceeds 35%, appropriately balancing the asymmetric costs. This threshold can be adjusted based on user preferences (risk-averse users might prefer $\tau^* = 0.25$, while autonomy-prioritizing users might prefer $\tau^* = 0.50$).

9 Research Agenda and Open Problems

Our analysis reveals several open research problems at the intersection of NLP, affective computing, human-computer interaction, and digital mental health.

9.1 Benchmark Datasets for Intent-Aware Emotion Understanding

Problem: Current sentiment analysis benchmarks focus on polarity accuracy, not appropriateness of system response. We need datasets including user text, context, goals, temporal sequences, and expert judgments on appropriate actions.

Challenges:

- Privacy: Emotional journal entries are highly sensitive
- Consent: Users may not want therapeutic interactions used for research
- Expert annotation: Requires clinical expertise
- Contextual richness: Need longitudinal data
- Diversity: Must represent different cultures, ages, contexts

Proposed Approach: Hybrid methodology combining simulated interactions, crowdsourced vignettes with expert validation, partnerships with digital mental health platforms (with consent), and synthetic data generation.

9.2 Interpretable Temporal Pattern Recognition

Problem: Deep learning time series models achieve high accuracy but lack interpretability. Users need to understand why the system identified a pattern.

Approaches:

- Attention mechanisms with visualization
- Symbolic pattern mining
- Hybrid neural-symbolic architectures
- Counterfactual explanations
- Case-based reasoning

9.3 Goal Inference from Implicit Signals

Problem: Users rarely state explicit goals. The system must infer goals from interaction patterns without over-interpreting limited data.

Formalization: Given history $\mathcal{H}$, estimate $P(g \mid \mathcal{H}; \theta)$ with confidence. Act on inferred goal only if $\max_g P(g \mid \mathcal{H}; \theta) > \tau$, otherwise prompt for clarification.

9.4 Pragmatic Reasoning in Affective Contexts

Problem: Emotional expression involves complex pragmatic phenomena (metaphor, hyperbole, indirect speech acts) requiring specialized treatment beyond standard pragmatic NLP.

9.5 Evaluation Metrics Beyond Accuracy

Problem: Standard metrics do not capture therapeutic appropriateness, user autonomy, or safety.

Proposed Metrics:

- User satisfaction and trust
- Alignment with clinician judgments
- False intervention rate vs. missed intervention rate
- Explanation quality
- Autonomy preservation
- Cross-cultural fairness

9.6 Cross-Cultural and Individual Variation

Problem: Emotional expression varies across cultures and individuals. Systems must adapt without stereotyping or requiring extensive personalization data.

Approaches:

- Personalized baseline estimation
- Multi-cultural training data
- Meta-learning for rapid adaptation
- Explicit uncertainty modeling in cross-cultural contexts

9.7 Human-AI Collaboration

Problem: Fully automated emotion interpretation may be neither achievable nor desirable.

Design Principles:

- Present uncertainty explicitly
- Offer multiple interpretations
- Enable user correction
- Learn from corrections
- Escalate to professionals when appropriate

9.8 Privacy-Preserving Personalization

Approaches:

- Federated learning (no centralized data)
- Differential privacy (noise for protection)
- On-device processing (local computation)
- User-controlled retention (automatic deletion)

9.9 Empirical Validation Strategy

To validate the proposed framework, we propose a controlled empirical study:

Study Design: Randomized comparison between sentiment-only and multi-layer intent-aware systems.

Participants: Users engaging in reflective journaling ($N \approx 100$, balanced across demographic groups).

Conditions:

- Baseline: Sentiment-only system (uses $S(x)$ for action selection)
- Treatment: Multi-layer framework (uses L_1–L_5 for action selection)

Primary Outcomes:

- False intervention rate: Percentage of inappropriate interventions (measured via user feedback and expert review)
- Missed support rate: Percentage of missed opportunities for helpful support (measured via expert review of interactions)
- User-perceived appropriateness: Likert-scale ratings of system responses

Secondary Outcomes:

- User trust (validated trust scale)
- System engagement (usage frequency, session duration)
- Explanation quality (user ratings of system explanations)
- Autonomy preservation (validated agency scale)

Analysis: Mixed-effects models comparing conditions on outcome measures, controlling for individual differences. Qualitative analysis of user feedback to identify nuanced failure modes.

Expected Result: Multi-layer system should demonstrate lower false intervention rate and higher appropriateness ratings while maintaining comparable missed support rate, validating the value of context, goals, and temporal modeling beyond sentiment alone.

This empirical design provides a concrete path toward validating the theoretical framework proposed in this paper.

10 Ethical and Privacy Considerations

The deployment of sentiment-aware systems in reflective and therapeutic contexts requires careful ethical consideration.

10.1 Informed Consent and Transparency

Users must understand what data is collected, how systems operate, what inferences may be drawn, and system limitations. Consent should be granular, allowing users to opt out of specific layers while retaining core functionality.

10.2 Privacy and Data Minimization

Systems should collect minimum necessary data, process locally when possible, encrypt emotional data, allow deletion without penalty, and implement retention limits.

Technical Approach: Federated learning enables pattern recognition while preserving privacy:

$$\text{Model}_{\text{global}} = \text{Aggregate}(\{\text{Model}_i + \mathcal{N}(0, \sigma^2 I)\}_{i=1}^{N}) \tag{65}$$

where noise σ is calibrated for (ϵ, δ)-differential privacy.

10.3 Avoiding Pathologization

Systems should normalize emotional variation and avoid diagnostic language. Negative sentiment does not equal mental health problems. Implementation requires normalization messages, avoiding clinical terminology, and contextualizing patterns.

10.4 Preserving User Autonomy

Default to minimal intervention. Offer support as options, not imperatives. Frame suggestions as questions. Respect user decisions even when system disagrees.

Example:

Poor: "You've been negative for 2 weeks. You should talk to someone."

Better: "I noticed you've been processing difficult feelings. Some find it helpful to talk with a friend or counselor. Would you like resources, or prefer to continue reflecting on your own?"

10.5 Accountability and Redress

Users should be able to correct system interpretations, receive explanations, and report harmful behavior. Developers must be accountable for systematic biases.

10.6 Professional Boundaries

Systems are not substitutes for professional mental health care. Must clearly state limitations, provide pathways to professional help, avoid medical advice, and not claim clinical efficacy without evidence.

When to Suggest Professional Help:

- Persistent negative sentiment (>3 weeks) with functional impairment
- Expression of suicidal ideation
- Description of harmful behaviors
- User explicitly requests help

Never:

- Diagnose conditions
- Prescribe treatments
- Substitute for emergency services

10.7 Regulatory Considerations

EU AI Act (Regulation EU 2024/1689) [6] classifies certain emotion recognition as high-risk, requiring risk management, data governance, technical documentation, transparency, and human oversight.

GDPR [5] treats emotional data as sensitive, requiring explicit consent, right to explanation, right to erasure, and data minimization.

10.8 Ethical Framework Summary

Core principles:

1. **Autonomy:** Respect user agency
2. **Beneficence:** Act in user's interests
3. **Non-Maleficence:** Avoid harm
4. **Justice:** Fair treatment across demographics
5. **Privacy:** Protect sensitive data
6. **Transparency:** Explain reasoning
7. **Accountability:** Accept responsibility

11 Conclusion

We provide a formal decision-theoretic foundation for intent-aware emotion modeling in reflective writing systems. Through formal analysis, we demonstrated that sentiment polarity $S(x)$ is not a sufficient statistic for determining appropriate system behavior, as optimal actions depend on conversational context, user goals, and temporal patterns.

The central contribution of this work is the utility-theoretic framing of emotional response selection. Specifically, our novel contributions include:

1. **Formal Insufficiency Proof:** We formally demonstrated that sentiment alone cannot determine appropriate actions in reflective contexts, requiring integration of context, goals, and temporal history.

2. **Asymmetric Cost Modeling:** We explicitly modeled the asymmetric costs of false intervention (C_{fp}: undermining autonomy) versus missed support (C_{fn}: failing to help), deriving optimal decision thresholds $\tau^* = C_{fp}/(C_{fp} + C_{fn})$.

3. **Ethical Constraints Formalization:** We formalized ethical constraints through hard action-space restrictions ($\mathcal{A}_{\text{allowed}}$) and soft utility penalties for privacy invasion and autonomy violation, contrasting with informal guidelines in prior work.

4. **Interpretable Utility Decomposition:** We decomposed utility into explicit components (appropriateness, safety, privacy, autonomy) with transparent weight specification methods, enabling inspection and adjustment unlike monolithic reward functions or implicit LLM objectives.

5. **Application to Open-Ended Reflective Writing:** We addressed open-ended emotional expression without predefined task goals, contrasting with task-oriented dialogue systems that optimize for completion metrics.

Observations from early-stage prototype deployment revealed systematic failure modes when sentiment-only approaches are applied: context collapse, therapeutic misalignment, intent ambiguity, privacy-invasive over-inference, and individual baseline variation. These failures stem from fundamental limitations in using sentiment as a proxy for user intent and therapeutic need.

The multi-layer intent modeling framework integrates sentiment detection with pragmatic reasoning, temporal pattern recognition, and goal-aware interpretation. This factorized architecture maintains interpretability through transparent layer outputs while enabling more appropriate responses than sentiment alone. The framework is not merely a modular dialogue policy but specifically addresses the unique challenges of reflective contexts: subjective appropriateness criteria, therapeutic considerations, and asymmetric intervention costs.

Positioning relative to existing work: Dialogue policy learning focuses on task completion; we address open-ended reflection. POMDP-based systems model state uncertainty; we additionally model explicit ethical constraints and asymmetric costs. RLHF learns preferences; we provide interpretable utility decomposition. LLMs can reason about context but lack calibrated uncertainty, transparent thresholds, and enforceable constraints that our framework provides.

The research agenda identifies concrete open problems requiring interdisciplinary collaboration: benchmark construction, interpretable temporal modeling, goal inference, evaluation metrics beyond accuracy, cross-cultural

adaptation, and privacy-preserving personalization. The proposed empirical validation strategy provides a path toward testing the framework through randomized comparison.

Ethical analysis emphasizes that technical sophistication must balance with respect for privacy, autonomy, and well-being. Informed consent, data minimization, avoiding pathologization, preserving autonomy, and maintaining professional boundaries are fundamental requirements. Regulatory frameworks including the EU AI Act and GDPR provide necessary constraints.

Limitations: This work presents a conceptual framework without full-scale empirical validation. The observations from prototype deployment are illustrative. Future work should implement the complete framework, conduct the proposed randomized study, develop standardized benchmarks, and investigate cross-cultural applicability and multimodal extensions.

As emotion-aware AI systems become prevalent in sensitive domains, the field must move toward holistic intent understanding integrating linguistic analysis with contextual reasoning, temporal dynamics, and ethical grounding. Our utility-theoretic framework provides foundation for systems respecting emotional complexity while offering meaningful support, balancing personalized assistance with privacy protection, and ensuring AI amplifies rather than replaces human judgment in emotional well-being.

Acknowledgments

I am grateful for insights from the NLP, affective computing, and digital mental health research communities that informed this work.

References

[1] Peter Brusilovsky and Eva Millán. User models for adaptive hypermedia and adaptive educational systems. In Peter Brusilovsky, Alfred Kobsa, and Wolfgang Nejdl, editors, *The Adaptive Web*, pages 3–53. Springer-Verlag, Berlin, 2007.

[2] Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, pages 4299–4307, 2017.

[3] Glen Coppersmith, Ryan Leary, Patrick Crutchley, and Alex Fine. Natural language processing of social media as screening for suicide risk. *Biomedical Informatics Insights*, 10:1178222618792860, 2018.

[4] Johannes C. Eichstaedt, Robert J. Smith, Raina M. Merchant, Lyle H. Ungar, Patrick Crutchley, Daniel Preoţiuc-Pietro, David A. Asch, and H. Andrew Schwartz. Facebook language predicts depression in medical records. *Proceedings of the National Academy of Sciences*, 115(44):11203–11208, 2018.

[5] European Parliament and Council of the European Union. Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). Official Journal of the European Union, 2016. OJ L 119/1. Available at: `https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32016R0679`.

[6] European Union. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, 2024. OJ L 2024/1689. Available at: `https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=OJ:L_202401689`.

[7] Emily G. Lattie, Elizabeth C. Adkins, Nathan Winquist, Colleen Stiles-Shields, Q. Eileen Wafford, and Andrea K. Graham. Digital mental health interventions for depression, anxiety, and enhancement of psychological well-being among college students: Systematic review. *Journal of Medical Internet Research*, 21(7):e12869, 2019.

[8] Stephen C. Levinson. *Pragmatics*. Cambridge University Press, Cambridge, UK, 1983.

[9] Feng Ma, Yao Yuan, Yi Xie, Huihui Ren, Ivan Liu, Meina Kan, Xilin Chen, Xiaohong Li, and Baochang Sun. Generative technology for human emotion recognition: A scoping review. *Information Fusion*, 113:102617, 2024.

[10] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, pages 27730–27744, 2022.

[11] Rosalind W. Picard. *Affective Computing*. MIT Press, Cambridge, MA, 1997.

[12] Kurt VanLehn. The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychologist*, 46(4):197–221, 2011.

[13] Jason D. Williams and Steve Young. Partially observable Markov decision processes for spoken dialog systems. *Computer Speech & Language*, 21(2):393–422, 2007.

[14] Steve Young, Milica Gašić, Blaise Thomson, and Jason D. Williams. POMDP-based statistical spoken dialog systems: A review. *Proceedings of the IEEE*, 101(5):1160–1179, 2013.

[15] Wenxuan Zhang, Yue Deng, Bing Liu, Sinno Jialin Pan, and Lidong Bing. Sentiment analysis in the era of large language models: A reality check. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 3881–3906, Mexico City, Mexico, 2024. Association for Computational Linguistics.